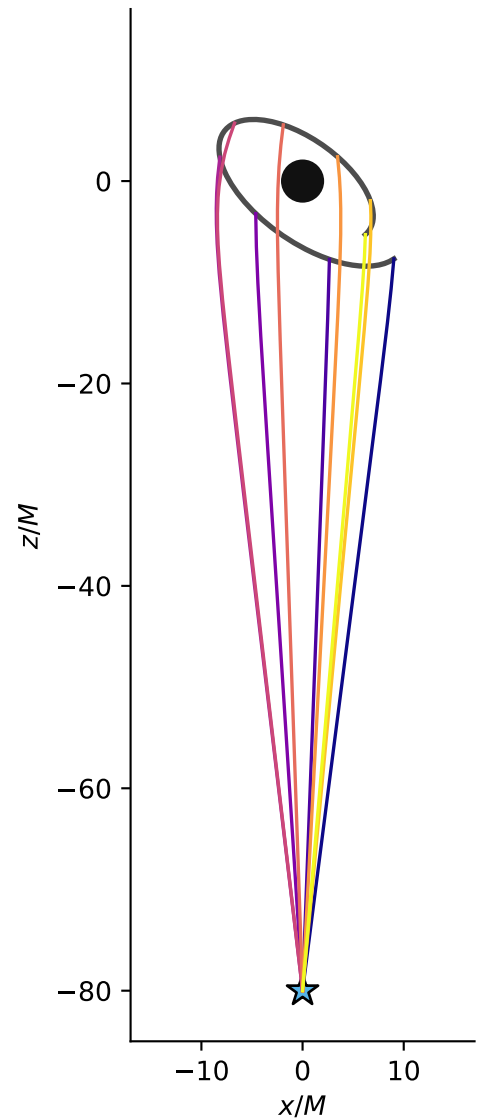
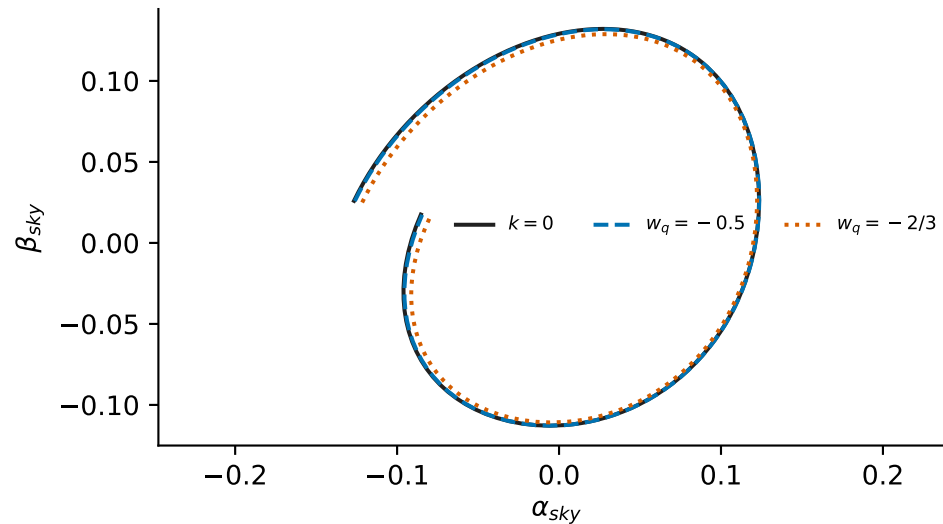


Physical geodesic shooting links spacetime parameters to complementary observables

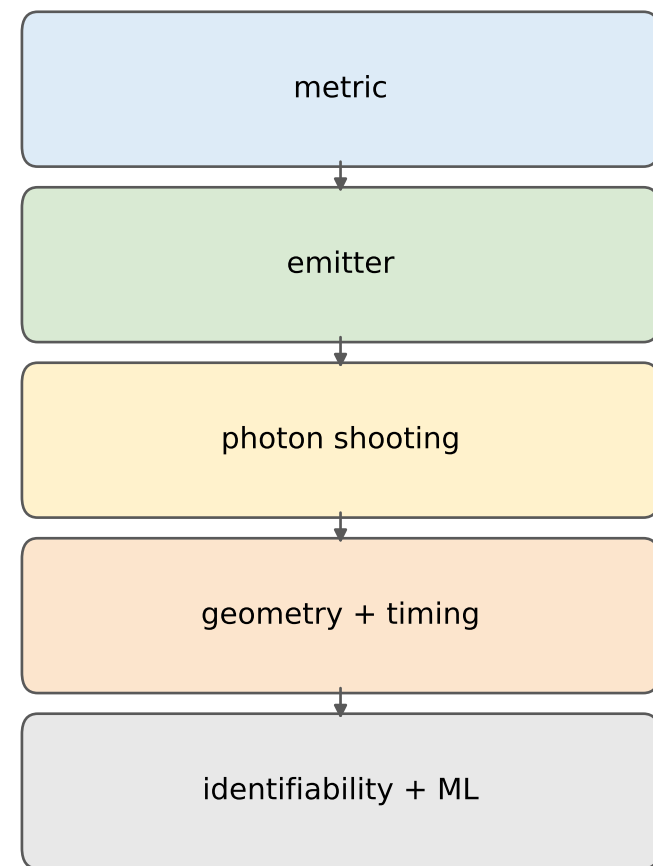
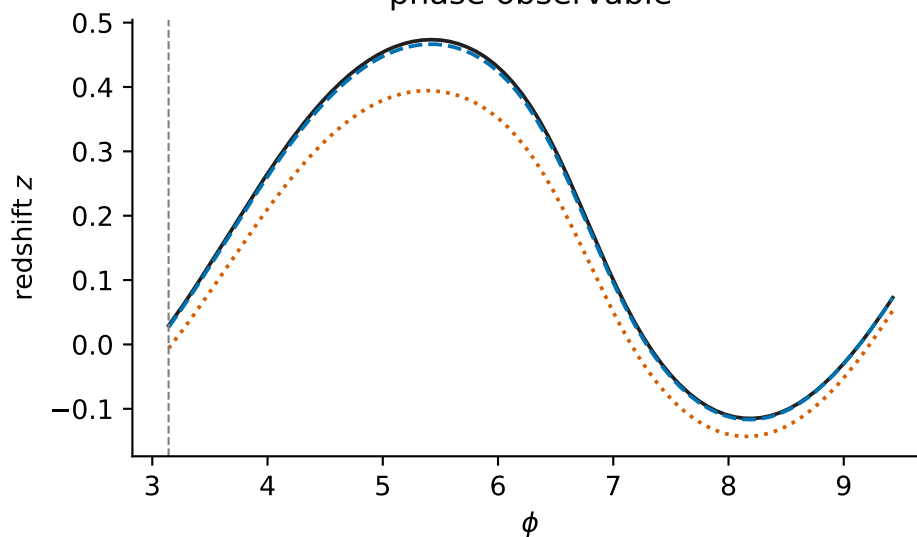
direct photon shooting



observer sky



phase observable



$k=0$: exact w_q rank loss